Analysis and Discussion

This document summarizes the analysis of NLP retrieval methods, potential LLM-based improvements, and ground truth considerations from the NLP Home Assignment.

Potential LLM Improvements for Retrieval

LLMs can enhance both lexical and semantic retrieval:

For Lexical Retrieval:

1. **Query/Document Expansion**: Use LLMs to add synonyms and related terms to queries or documents, bridging the lexical gap and improving TF-IDF performance.
2. **Re-ranking**: Apply LLMs to re-rank initial lexical results (e.g., from TF-IDF) based on deeper semantic relevance.

For Semantic Search:

1. **Hybrid Retrieval**: Combine lexical and semantic scores, potentially using LLMs to determine optimal weighting based on query analysis.
2. **Query Understanding & Reformulation**: Leverage LLMs to interpret complex user intent and rewrite queries for better system matching.
3. **Enhanced Embeddings**: Generate context-aware embeddings that capture narrative roles or fine-tune embeddings on domain-specific data for increased relevance.

General Enhancements:

1. **Relevance Feedback**: Train LLMs on user interactions to iteratively refine retrieval strategies.
2. **Multi-hop Reasoning**: Enable answering complex queries by having LLMs synthesize information from multiple retrieved passages.
3. **Explainability**: Use LLMs to generate explanations for retrieved results, increasing user trust.

Drawbacks and Limitations of LLMs in Retrieval

Using LLMs also introduces challenges:

1. **Computational Cost**: High resource requirements and potential latency issues.
2. **Hallucination Risk**: LLMs might generate incorrect information, adding noise if used for expansion.
3. **Training Data Bias**: Inherited biases can lead to skewed or unfair results.

4. **Lack of Transparency**: The "black box" nature makes debugging difficult.
5. **Contextual Limits**: Finite token windows restrict processing of very long texts.
6. **Evaluation Challenges**: Assessing true semantic relevance often requires costly human judgment.

Creating a Reliable Ground Truth for Evaluation

Relying on one model's output (like miniLM) as ground truth is suboptimal. A better approach involves:

1. Human Annotation

- **Core Idea**: Use multiple human experts to judge document relevance for predefined queries. This is the gold standard.

2. Robust Annotation Process

- **Key Elements**:
  - Define diverse query types (factual, thematic, etc.).
  - Use a graded relevance scale (e.g., 0-3) instead of binary judgments.
  - Ensure inter-annotator agreement (IAA) through metrics and disagreement resolution.

3. Efficiency via Pooling

- **Method**: Run multiple retrieval systems, collect the top-k results from each into a pool, and have humans annotate only the documents in this pool. Assumes non-pooled documents are irrelevant.

4. Comprehensive Evaluation Design

- **Considerations**: Account for narrative context beyond single chunks, use query variations to test robustness, and ensure a balanced set of queries covering different aspects and difficulties.

Implementing a human-centric annotation process, possibly using pooling, yields a more reliable ground truth for evaluating and comparing retrieval methods.